

# **Lecture 03a – $t$ -Tests**

ENVX2001 Applied Statistical Methods

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## The $t$ -test (revision)

## Introduction

- **One-sample**  $t$ -test: compare a sample mean against a known value
- **Two-sample**  $t$ -test: compare the means of two independent groups
- **Paired**  $t$ -test: compare measurements taken on the same subjects (e.g. before and after)

Last week we learned about **confidence intervals**, which estimate the uncertainty of a parameter (e.g. a mean).

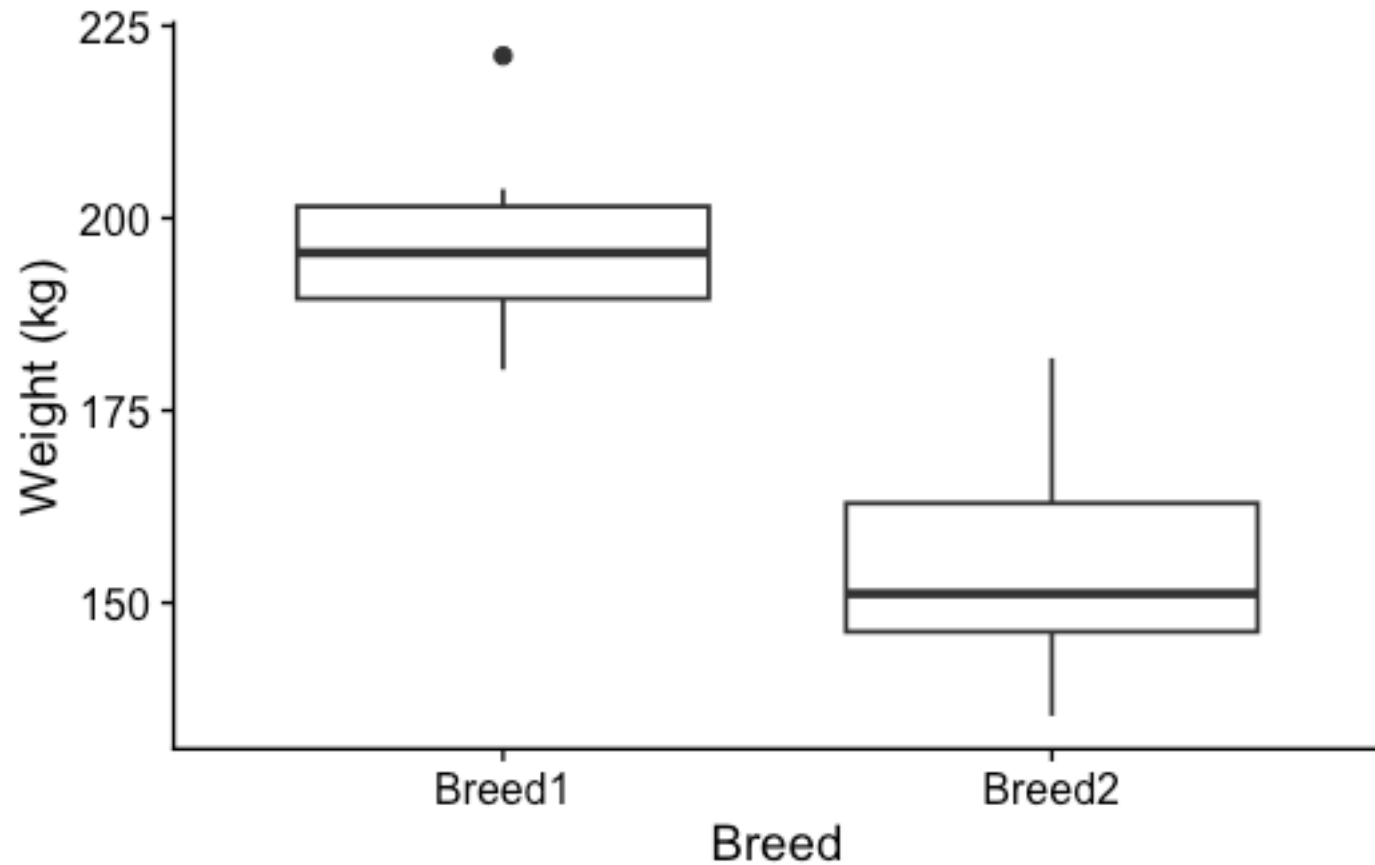
The  $t$ -test and the confidence interval are closely related. Both are derived from the same quantities, and this week we look at that connection more carefully.

## Cattle

Two breeds of cattle, 12 and 15 samples. Are the breeds different in mean weight?

```
cattle ← read.csv("data/cattle.csv") |>
  pivot_longer(cols = everything(), names_to = "breed", values_to = "weight") |>
  mutate(breed = as.factor(breed)) |>
  drop_na()

ggplot(cattle, aes(breed, weight)) +
  geom_boxplot() +
  labs(x = "Breed", y = "Weight (kg)") +
  cowplot::theme_cowplot()
```



The boxplot suggests a difference. To test, you would run `t.test()` and check the p-value. **What is the test actually computing?**

## From CIs to the $t$ -test

## The difference between two means

In the last lecture, we constructed confidence intervals for a single mean. The same principle extends to the **difference** between two means.

- Estimate the difference:  $\bar{y}_1 - \bar{y}_2$
- Construct a 95% confidence interval around that estimate
- Evaluate whether the interval includes zero
- If the interval **excludes zero**: the means are significantly different at the 5% level
- If the interval **includes zero**: the data are consistent with no difference

## `t.test()` gives you both

```
t.test(weight ~ breed, data = cattle, var.equal = TRUE)
```

### Two Sample t-test

data: weight by breed

t = 9.4624, df = 25, p-value = 9.663e-10

alternative hypothesis: true difference in means between group Breed1 and group Breed2 is not equal to 0

95 percent confidence interval:

33.23011 51.71989

sample estimates:

| mean in group Breed1 | mean in group Breed2 |
|----------------------|----------------------|
| 196.175              | 153.700              |

The output reports two quantities:

- A **p-value** (probability of observing a result this extreme if the null hypothesis were true)
- A **95% confidence interval** for the difference between the two means



These always agree. If the CI excludes zero, the p-value is below 0.05, and vice versa. The  $t$ -statistic measures how far the observed difference is from zero, relative to its standard error.

## Exploring the confidence interval

```
{ojs}  
//| echo: false  
import { ciDifferenceWidget } from "../assets/js/ttest-widget.js"  
ciDifferenceWidget()
```

- Increasing the difference between means moves the CI away from zero
- Decreasing variability narrows the CI
- When the interval excludes zero, the means are significantly different

## The CI for a difference

In the last lecture, we built a 95% CI for a single mean:

$$\bar{y} \pm t^* \times SE$$

For the **difference** between two means, the same structure applies:

$$(\bar{y}_1 - \bar{y}_2) \pm t^* \times SE(\bar{y}_1 - \bar{y}_2)$$

If this interval excludes zero, the groups differ significantly.

## The $t$ -statistic

We can rearrange the CI formula to isolate a single number that captures both the difference and the variability:

$$t = \frac{\bar{y}_1 - \bar{y}_2}{SE(\bar{y}_1 - \bar{y}_2)}$$

The  $t$ -statistic counts how many standard errors the observed difference is from zero.

For the cattle data:  $t = 42.5/4.5 = 9.46$ . The observed difference is over nine standard errors from zero.

## From the $t$ -statistic to the p-value

We have a  $t$ -statistic. The p-value answers: if there were truly no difference between the breeds, how often would we see a value of  $|t|$  this large or larger?

- A small p-value means the observed difference is hard to explain by chance alone
- A large p-value means the observed difference is consistent with random variation

The 95% CI and the p-value always agree. If the 95% CI excludes zero, the p-value is below 0.05, and vice versa.

## Exploring the p-value

```
{ojs}  
//| echo: false  
import { pvalueExplorerWidget } from "../assets/js/ttest-widget.js"  
pvalueExplorerWidget()
```

- The shaded area under both tails is the p-value
- As the  $t$ -statistic moves further from zero, the p-value decreases
- At low degrees of freedom the tails are heavier, but the shape approaches the normal as df increases

# **The hypothesis testing workflow**

## Step 1: State what we are testing

- **Null hypothesis:**  $H_0 : \mu_1 = \mu_2$  (no difference in mean weight between breeds)
- **Alternative hypothesis:**  $H_1 : \mu_1 \neq \mu_2$  (means differ)
- This is a **two-tailed** test: we are looking for a difference in either direction (Breed 1 heavier *or* lighter than Breed 2), not just one

**Model equation:**  $y_{ij} = \mu_i + \varepsilon_{ij}$

Each observation is the group mean plus random error. The hypothesis asks whether  $\mu_1$  and  $\mu_2$  are equal.



## Step 2a: Equal variances

The  $t$ -test assumes both groups have similar variability. If they do not, then the test is unreliable because the pooled estimate of variance will not represent either group well.

- Rule of thumb: the ratio of the larger SD to the smaller SD should be below 2.0

```
cattle_summary ← cattle |>
  group_by(breed) |>
  summarise(n = n(), mean_wt = mean(weight), sd_wt = sd(weight))
```

```
cattle_summary
```

```
# A tibble: 2 × 4
  breed      n mean_wt sd_wt
  <fct> <int>   <dbl> <dbl>
1 Breed1    12    196.   10.6
2 Breed2    15    154.   12.3
```

```
max(cattle_summary$sd_wt) / min(cattle_summary$sd_wt)
```

[1] 1.158756

## Step 2b: Normality

The  $t$ -test assumes the data in each group follow a normal distribution. With small samples, departures from normality (strong skew, heavy tails, outliers) can distort the p-value and confidence interval. Check each group separately.

```
shapiro.test(cattle$weight[cattle$breed == "Breed1"])
```

Shapiro-Wilk normality test

```
data:  cattle$weight[cattle$breed == "Breed1"]  
W = 0.94084, p-value = 0.5091
```

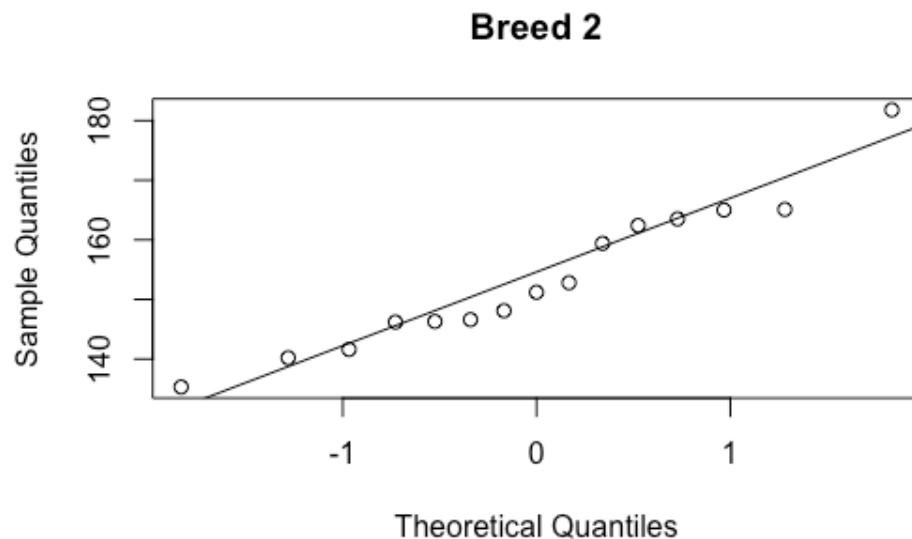
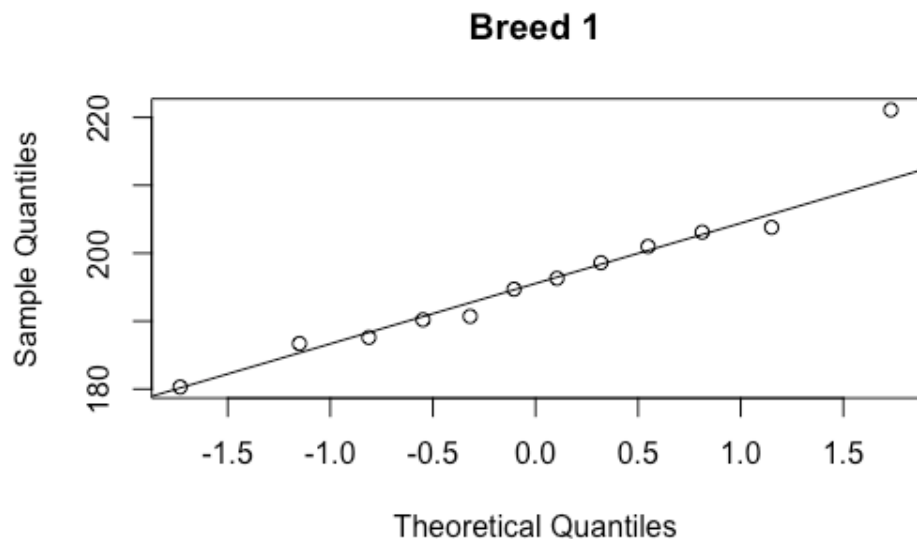
```
shapiro.test(cattle$weight[cattle$breed == "Breed2"])
```

Shapiro-Wilk normality test

```
data:  cattle$weight[cattle$breed == "Breed2"]  
W = 0.94636, p-value = 0.4691
```

## Step 2b: Normality (QQ plots)

```
qqnorm(cattle$weight[cattle$breed == "Breed1"], main = "Breed 1")  
qqline(cattle$weight[cattle$breed == "Breed1"])  
qqnorm(cattle$weight[cattle$breed == "Breed2"], main = "Breed 2")  
qqline(cattle$weight[cattle$breed == "Breed2"])
```



If points fall close to the line, the data are approximately normal. Both groups pass the Shapiro-Wilk test ( $p > 0.05$ ) and the QQ plots show no strong departures.

## Step 2c: Independence

Observations should not influence each other. This is a property of the study design, not something we test statistically.

- Each animal is measured once
- The weight of one animal does not affect the weight of another
- If the same animals were measured before and after a treatment, observations would not be independent (use a paired  $t$ -test instead)

### Step 3: Run the test and interpret

```
t.test(weight ~ breed, data = cattle, var.equal = TRUE)
```

Two Sample t-test

data: weight by breed

t = 9.4624, df = 25, p-value = 9.663e-10

alternative hypothesis: true difference in means between group Breed1 and group Breed2 is not equal to 0

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sample estimates:

mean in group Breed1 mean in group Breed2

196.175

153.700

- The  $t$ -statistic, degrees of freedom ( $n_1 + n_2 - 2$ ), and p-value are reported together
- `var.equal = TRUE` performs the pooled (Student's)  $t$ -test, which assumes equal variances

A p-value of 0.03 does not mean there is a 3% chance the null hypothesis is true. It means: if there were truly no difference, we would observe a result this extreme about 3% of the time.



## HATPC

Notice the five steps we just followed:

1. **H**ypotheses: state the null and alternative
2. **A**ssumptions: check equal variances, normality, independence
3. **T**est: run the appropriate test
4. **P**-value: interpret the result
5. **C**onclusion: state the finding in context

We will use this framework throughout the semester, starting with ANOVA in the next lecture.

## What if assumptions are not met?

- **Unequal variances:** use Welch's  $t$ -test (the default in R when `var.equal = FALSE`)
- **Non-normality:** if  $N > 30$ , the Central Limit Theorem provides reasonable coverage
- **Non-independence:** consider a **paired  $t$ -test** (e.g. before and after measurements on the same subjects)

# Summary

## Key points

- The **two-sample  $t$ -test** compares means of two independent groups
- The  $t$ -test and the CI for the difference are two views of the same calculation
- Check the three assumptions (equal variances, normality, independence) before running any test
- Follow **HATPC** as a structured process for hypothesis testing

In Lab 03, you will practise this workflow with `t.test()` and diagnostic plots.

**Next:** what if we have more than two groups? We need **ANOVA**.

## 15 minute break

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